Lab 2: Improving Agent Quality through AI Instructions and Example Queries

Learning Objectives

In this lab, you will build a Fabric Data Agent that answers questions using the ecommerce lakehouse and configure it for improved responses. You'll start by creating the agent and exploring the lakehouse data through a few sample prompts while inspecting the run steps (paraphrasing, SQL generation, and query output). Next, you'll add Agent Instructions and Data source instructions to guide how the agent asks clarifying questions and formats responses. Finally, you'll use example queries to produce more consistent results.

Pre-requisites

1. You have completed Lab 1 – Getting Started

Lab Steps

Step 1: Data exploration through Data Agent

At the end of Lab 1, you asked the Data Agent - What are the top 5 rows in customers table?

1. Let's start by understanding the response structure for the Data Agent.

Here are the top 5 rows from the Customers table:

1. Customer ID: 0000366f3b9a7992bf8c76cfdf3221e2, Zip Code: 7787, City: CAJAMAR, State: SP
2. Customer ID: 0000b849f77a49e4a4ce2b2a4ca5be3f, Zip Code: 6053, City: OSASCO, State: SP
3. Customer ID: 0000f46a3911fa3c0805444483337064, Zip Code: 88115, City: SAO JOSE, State: SC
4. Customer ID: 0000f6ccb0745a6a4b88665a16c9f078, Zip Code: 66812, City: BELEM, State: PA
5. Customer ID: 0004aac84e0df4da2b147fca70cf8255, Zip Code: 18040, City: SOROCABA, State: SP

Let me know if you need more details or a different view!

1 step completed

Response time: 16 sec

Data agent responds by taking the values from the table and outputs 5 rows from the table.

We can also take this further and view the steps by clicking further on the step to expand the response:

1 step completed

Response time: 14 sec

Analyzed ecommerce_lakehouse Lakehouse for: "Show the top 5 rows from the customers table."

1 step completed

Response time: 16 sec

Analyzed ecommerce_lakehouse Lakehouse for: "Show the first 5 rows from the Customers table with all available columns."

Query code

Sql

```sql
SELECT TOP 5 *
FROM [fda].[Customers];
```

Query output

Customer ID	Customer Zip Code	Customer City	Customer State
0000366f3b9a7992bf8c76cfdf3221e2	7787	CAJAMAR	SP
0000b849f77a49e4a4ce2b2a4ca5be3f	6053	OSASCO	SP
0000f46a3911fa3c0805444483337064	88115	SAO JOSE	SC
0000f6ccb0745a6a4b88665a16c9f078	66812	BELEM	PA
0004aac84e0df4da2b147fca70cf8255	18040	SOROCABA	SP

This should give you an SQL query that was used to generate the response. These can be used by the user to understand what steps Data Agent took to generate a response.

2. Now you can try asking some additional data exploration questions.
   Take this time to understand the results of the query and how Data Agent came up with the result. Ask the following questions to start:

a. How many unique cities are sellers from?

Analyzed ecommerce_lakehouse Lakehouse for: "How many unique cities are sellers from in the Sellers table?"

Query code

Sql

```sql
SELECT COUNT(DISTINCT [Seller City]) AS UniqueSellerCities
FROM [fda].[Sellers];
```

Query output

UniqueSellerCities
611

b. What is the most common product category purchased?

Analyzed ecommerce_lakehouse Lakehouse for: "What is the most common product category purchased?"

Query code

Sql

```sql
SELECT TOP 1
    p.[Product Category],
    COUNT(*) AS PurchaseCount
FROM [fda].[Sales] s
INNER JOIN [fda].[Products] p
    ON s.[Product ID] = p.[Product ID]
GROUP BY p.[Product Category]
ORDER BY COUNT(*) DESC
```

Query output

Product Category	PurchaseCount
cama_mesa_banho	11115

c. What was the most expensive order purchased to date?

Analyzed ecommerce_lakehouse Lakehouse for: "What was the most expensive order purchased to date? Provide the order ID and the total amount."

Query code

Sql

```sql
SELECT TOP 1
    [Order ID],
    SUM([Line Total]) AS [Total Amount]
FROM
    [fda].[Sales]
GROUP BY
    [Order ID]
ORDER BY
    [Total Amount] DESC
```

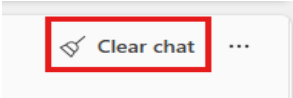
Query output

Order ID	Total Amount
03caa2c082116e1d31e67e9ae3700499	13664.08

3. You're now ready to move on to the next step in this lab.

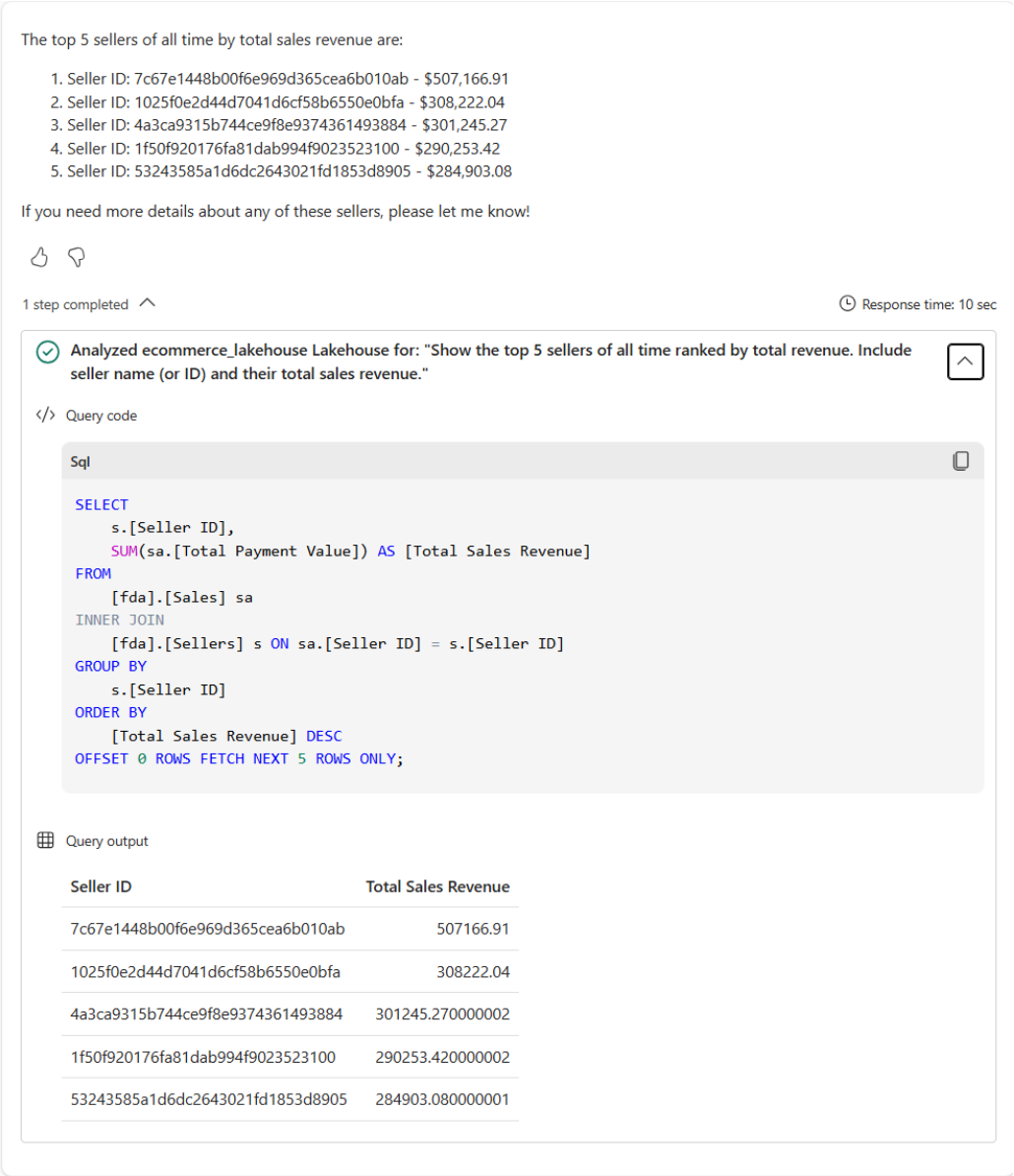
Step 2: Improving Results Using Agent Instructions

1. Let's start by clicking "Clear Chat" on the top right corner.



2. Ask your agent the following question: **Who are the top 5 sellers of all time by revenue?**

   You may see a response like the below screenshot:



The top 5 sellers of all time by total sales revenue are:

1. Seller ID: 7c67e1448b00f6e969d365cea6b010ab - $507,166.91
2. Seller ID: 1025f0e2d44d7041d6cf58b6550e0bfa - $308,222.04
3. Seller ID: 4a3ca9315b744ce9f8e9374361493884 - $301,245.27
4. Seller ID: 1f50f920176fa81dab994f9023523100 - $290,253.42
5. Seller ID: 53243585a1d6dc2643021fd1853d8905 - $284,903.08

If you need more details about any of these sellers, please let me know!

1 step completed

Response time: 10 sec

Analyzed ecommerce_lakehouse Lakehouse for: "Show the top 5 sellers of all time ranked by total revenue. Include seller name (or ID) and their total sales revenue."

Query code

Sql

```sql
SELECT
    s.[Seller ID],
    SUM(sa.[Total Payment Value]) AS [Total Sales Revenue]
FROM
    [fda].[Sales] sa
INNER JOIN
    [fda].[Sellers] s ON sa.[Seller ID] = s.[Seller ID]
GROUP BY
    s.[Seller ID]
ORDER BY
    [Total Sales Revenue] DESC
OFFSET 0 ROWS FETCH NEXT 5 ROWS ONLY;
```

Query output

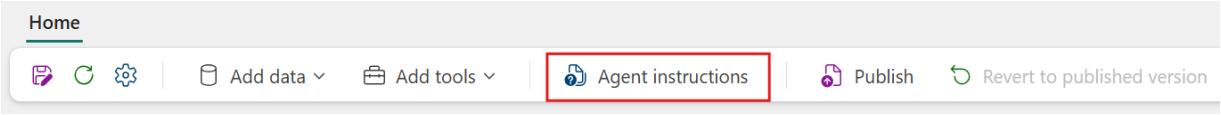
Seller ID	Total Sales Revenue
7c67e1448b00f6e969d365cea6b010ab	507166.91
1025f0e2d44d7041d6cf58b6550e0bfa	308222.04
4a3ca9315b744ce9f8e9374361493884	301245.270000002
1f50f920176fa81dab994f9023523100	290253.420000002
53243585a1d6dc2643021fd1853d8905	284903.080000001

The agent provides a response by analyzing all the sales data based on revenue in a table format with minimal details.

We can improve the agent's response by instructing it to respond with descriptive fields.

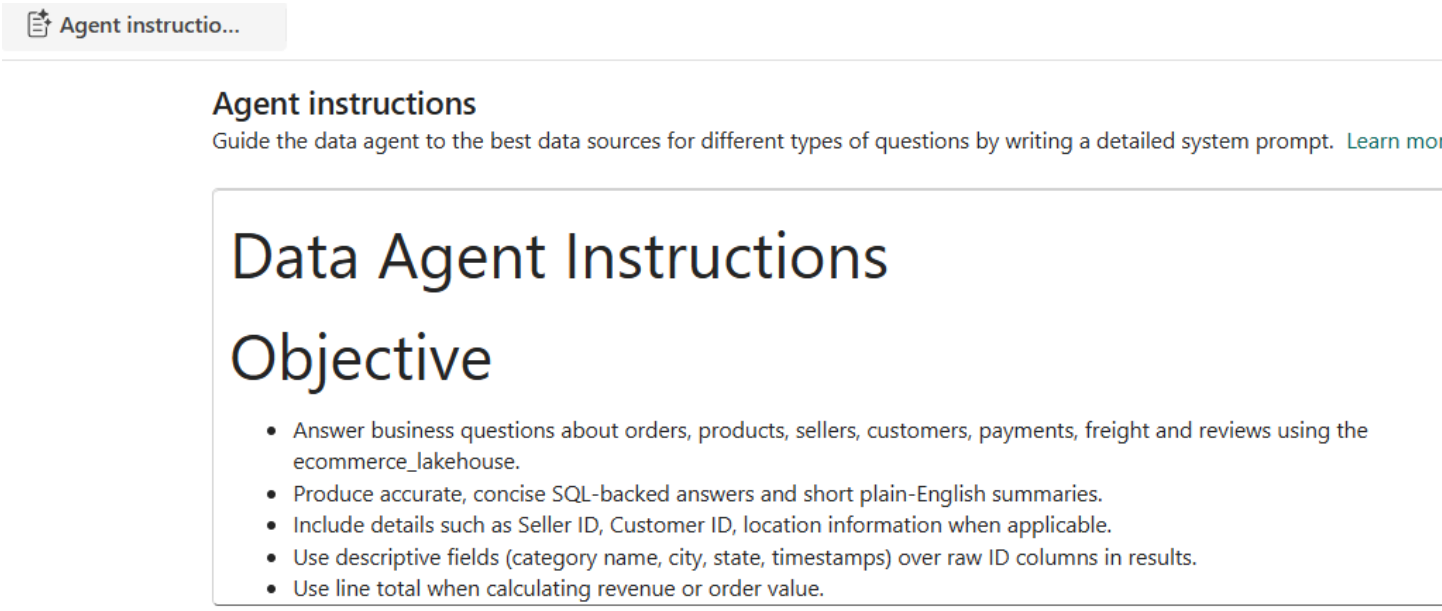
Let's also include instructions to add a short, summarized answer to the question to help users understand the results more quickly.

3. Select "Agent Instructions" In the Ribbon

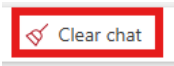


4. Add the following instructions to the Agent:

```
# Data Agent Instructions
# Objective
- Answer business questions about orders, products, sellers, customers, payments, freight and reviews using the ecommerce_lakehouse.
- Produce accurate, concise SQL-backed answers and short plain-English summaries.
- Include details such as Seller ID, Customer ID, location information when applicable.
- Use descriptive fields (category name, city, state, timestamps) over raw ID columns in results.
- Use line total when calculating revenue or order value.
```



You may now close the tab for Agent instructions



5. Click “Clear Chat” on the top right corner.
6. Try asking Data Agent the same question again – **Who are the top 5 sellers of all time by revenue?**

   Notice that the agent now responds according to the instructions added:

The top 5 sellers of all time by revenue are:

1. Seller ID: 4869f7a5dfa277a7dca6462dcf3b52b2 - Total Revenue: 249,640.70 - Number of Orders: 1,132
2. Seller ID: 7c67e1448b00f6e969d365cea6b010ab - Total Revenue: 239,536.44 - Number of Orders: 982
3. Seller ID: 53243585a1d6dc2643021fd1853d8905 - Total Revenue: 235,856.68 - Number of Orders: 358
4. Seller ID: 4a3ca9315b744ce9f8e9374361493884 - Total Revenue: 235,539.96 - Number of Orders: 1,806
5. Seller ID: fa1c13f2614d7b5c4749cbc52fecda94 - Total Revenue: 204,084.73 - Number of Orders: 585

These sellers have generated the highest total revenue on the platform.

1 step completed

Response time: 10 sec

Analyzed ecommerce_lakehouse Lakehouse for: "List the top 5 sellers of all time ranked by total revenue (sum of line total), including their Seller ID, total revenue, and number of orders."

Analyzed ecommerce_lakehouse Lakehouse for: "List the top 5 sellers of all time ranked by total revenue (sum of line total), including their Seller ID, total revenue, and number of orders."

Query code

Sql

```sql
SELECT
    s.[Seller ID],
    SUM(s.[Line Total]) AS TotalRevenue,
    COUNT(DISTINCT s.[Order ID]) AS NumberOfOrders
FROM
    [fda].[Sales] s
GROUP BY
    s.[Seller ID]
ORDER BY
    TotalRevenue DESC
OFFSET 0 ROWS FETCH NEXT 5 ROWS ONLY
```

Clear chat

7. Click “Clear chat” on the top right corner of the screen.

8. When you try asking Data Agent any of the questions below, you’re likely to receive follow-ups that will allow you to explore different filters.
   (Tip: Clear chat between your questions)
   a. **What were the top 3 products sold over the most recent six months?**

The top 3 products sold over the most recent six months were:

1. Watches & Gifts (Product ID: 53b36df67ebb7c41585e8d54d6772e08) with 250 units sold.
2. Furniture & Decor (Product ID: aca2eb7d00ea1a7b8ebd4e68314663af) with 245 units sold.
3. Computers & Accessories (Product ID: 3dd2a17168ec895c781a9191c1e95ad7) with 181 units sold.

1 step completed

Response time: 11 sec

Analyzed ecommerce_lakehouse Lakehouse for: "Identify the top 3 best-selling products by quantity in the most recent six months available. Include product category (English) and the total quantity sold."

Query code

Sql

```sql
WITH RecentMonths AS (
    SELECT TOP 6
        [Order Year],
        [Order Month]
    FROM [fda].[Sales]
    WHERE [Order Status] = 'delivered'
    GROUP BY [Order Year], [Order Month]
    ORDER BY [Order Year] DESC, [Order Month] DESC
),
RecentSales AS (
    SELECT s.*
    FROM [fda].[Sales] s
    INNER JOIN RecentMonths rm
        ON s.[Order Year] = rm.[Order Year]
        AND s.[Order Month] = rm.[Order Month]
    WHERE s.[Order Status] = 'delivered'
)
SELECT TOP 3
    p.[Product ID],
    p.[Product Category English],
    COUNT(*) AS [Total Quantity Sold]
FROM RecentSales rs
INNER JOIN [fda].[Products] p
    ON rs.[Product ID] = p.[Product ID]
GROUP BY p.[Product ID], p.[Product Category English]
ORDER BY [Total Quantity Sold] DESC
```

Query output

Product ID	Product Category English	Total Quantity Sold
53b36df67ebb7c41585e8d54d6772e08	watches_gifts	250
aca2eb7d00ea1a7b8ebd4e68314663af	furniture_decor	245
3dd2a17168ec895c781a9191c1e95ad7	computers_accessories	181

b. **Which state had the highest sales in the last year?**

The state with the highest total sales in the last year was São Paulo (SP), with total sales amounting to 2,429,203.94.

1 step completed

Response time: 11 sec

Analyzed ecommerce_lakehouse Lakehouse for: "Identify the state with the highest total sales in the most recent full year available, based on line total from the sales data."

Query code

Sql

```sql
WITH MostRecentFullYear AS (
    SELECT MAX([Order Year]) AS RecentYear
    FROM [fda].[Sales]
    WHERE [Order Status] = 'delivered'
      AND [Order Year] IS NOT NULL
      AND [Order Year] < (SELECT YEAR(GETDATE()))
      AND [Order Year] IN (
            SELECT [Order Year]
            FROM [fda].[Sales]
            GROUP BY [Order Year]
            HAVING COUNT(DISTINCT [Order Month]) = 12
        )
)
SELECT TOP 1
    c.[Customer State],
    SUM(s.[Line Total]) AS TotalSales
FROM [fda].[Sales] s
INNER JOIN [fda].[Customers] c
    ON s.[Customer ID] = c.[Customer ID]
CROSS JOIN MostRecentFullYear y
WHERE s.[Order Year] = y.RecentYear
  AND s.[Order Status] = 'delivered'
GROUP BY c.[Customer State]
ORDER BY TotalSales DESC
```

Query output

Customer State	TotalSales
SP	2429203.94000003

c. **What city has the greatest number of unique customers?**

The city with the greatest number of unique customers is São Paulo, with 14,970 unique customers.

1 step completed

Response time: 8 sec

Analyzed ecommerce_lakehouse Lakehouse for: "Which city has the greatest number of unique customers? Provide the city name and the corresponding count of unique customers."

Query code

Sql

```sql
SELECT TOP 1
    [Customer City] AS City,
    COUNT(DISTINCT [Customer ID]) AS UniqueCustomerCount
FROM [fda].[Customers]
GROUP BY [Customer City]
ORDER BY UniqueCustomerCount DESC;
```

Query output

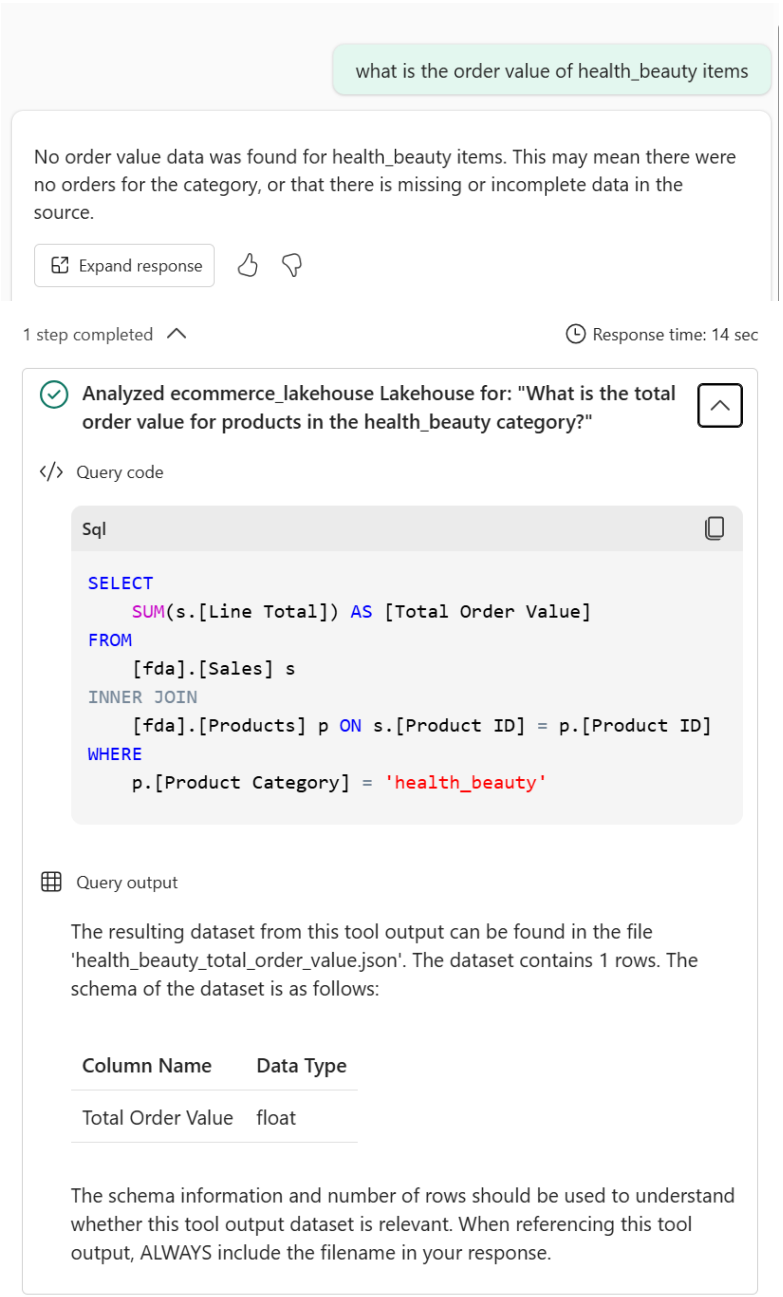
City	UniqueCustomerCount
SAO PAULO	14970

Step 3: Improving Results Using Data Source Instructions

1. Let's start by clicking "Clear Chat".

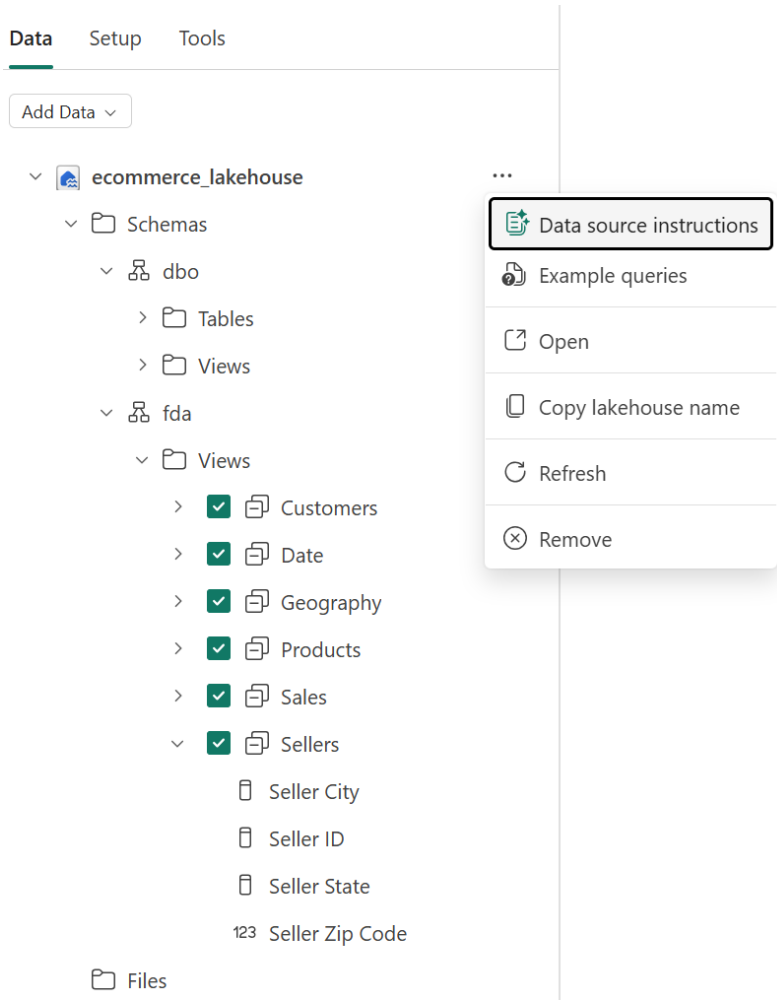
2. Ask the question: **What is the order value of health_beauty items?**

It is unlikely that Data Agent will know all the values in the table. Hence, the Data Agent is unable to determine how to accurately create a SQL query.



3. To help the Data Source with this question, let's add some Data Source instructions:

   Click on the three dots next to the ecommerce_lakehouse and click on data source instructions.



4. Add the following Instructions in your data source instructions:

Table Descriptions
- The Products table contains Product Id, Product description length, product name length, product dimensions (product height, product length, and product width), product photos count, product weight. Products table also has columns for Product Category (in Portuguese) and

Product Category English which are English translations of Product Category. Values in Product Category English may describe items like electronics, home goods, apparel, furniture etc.
- The customers table includes customer ID and geolocation information such as their city, state and zip code.

Data source description

Describe what's in this data source, what it represents, and why someone might use it to answer a question. Learn more

Enter your text in markdown format.

Data source instructions

For each connected data source, help the data agent understand its data and how to use it most effectively. Learn more

Format: Markdown

```
## Table Descriptions
- The Products table contains Product Id, Product description length, product name length, product dimensions (product height, product length, and product width), product photos count, product weight. Products table also has columns for Product Category (in Portuguese) and Product Category English which are English translations of Product Category. Values in Product Category English may describe items like electronics, home goods, apparel, furniture etc.
- The customers table includes customer ID and geolocation information such as their city, state and zip code.
```

595/15000 characters used

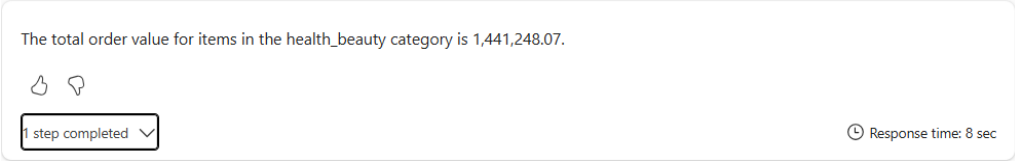
Write some markdown to preview

Type text in markdown format in the text area to see a live preview here.

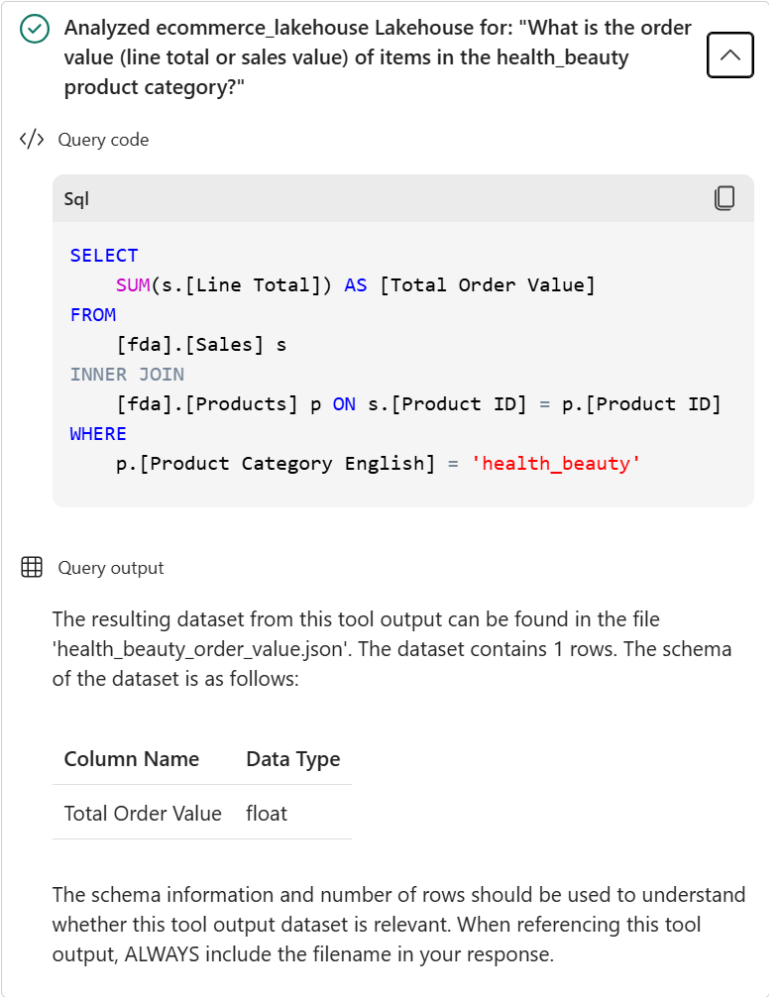
5. Let’s clear the chat, and try asking the question again: **What is the**

order value of health_beauty items?

Did you notice a difference in the response this time?



Expand the agent’s response to view how your instructions impacted the query.



6. Other questions you may try:

- Give me the revenue from furniture related purchases?

The total revenue from furniture-related purchases is 1,420,916.52.

1 step completed

Response time: 10 sec

Analyzed ecommerce_lakehouse Lakehouse for: "What is the total revenue from purchases of furniture and related products, considering all orders and the product categories relevant to furniture?"

Query code

Sql

```sql
SELECT
    SUM(s.[Line Total]) AS TotalFurnitureRevenue
FROM
    [fda].[Sales] s
INNER JOIN
    [fda].[Products] p ON s.[Product ID] = p.[Product ID]
WHERE
    LOWER(p.[Product Category English]) LIKE '%furniture%'
```

Query output

TotalFurnitureRevenue
1420916.52

- How many items were purchased for kids?

How many items were purchased for kids?

A total of 7,190 items were purchased in categories related to kids, such as toys, kids, or baby products.

1 step completed

Response time: 10 sec

Analyzed ecommerce_lakehouse Lakehouse for: "Count the number of products purchased in categories related to kids or children, such as 'toys', 'kids', 'baby', or similar, and provide the total item count."

Query code

Sql

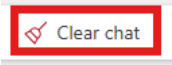
```sql
SELECT COUNT(*) AS TotalItemCount
FROM [fda].[Sales] s
INNER JOIN [fda].[Products] p ON s.[Product ID] = p.[Product ID]
WHERE
    p.[Product Category English] IS NOT NULL AND (
        LOWER(p.[Product Category English]) LIKE '%toy%' OR
        LOWER(p.[Product Category English]) LIKE '%kid%' OR
        LOWER(p.[Product Category English]) LIKE '%child%' OR
        LOWER(p.[Product Category English]) LIKE '%baby%'
    )
```

Query output

TotalItemCount
7190

Step 4: Improving Results Using Example Queries

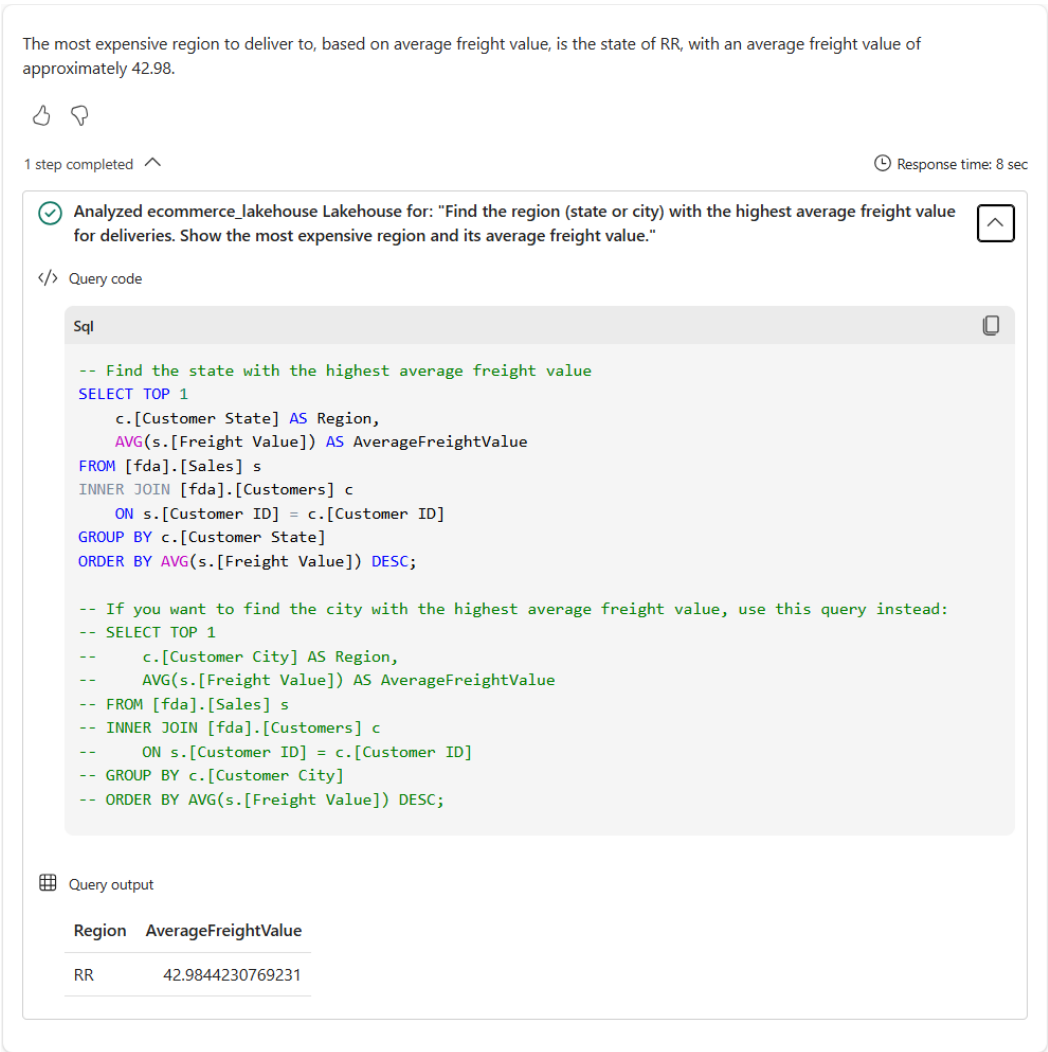
1. Let’s start by clicking “Clear Chat”



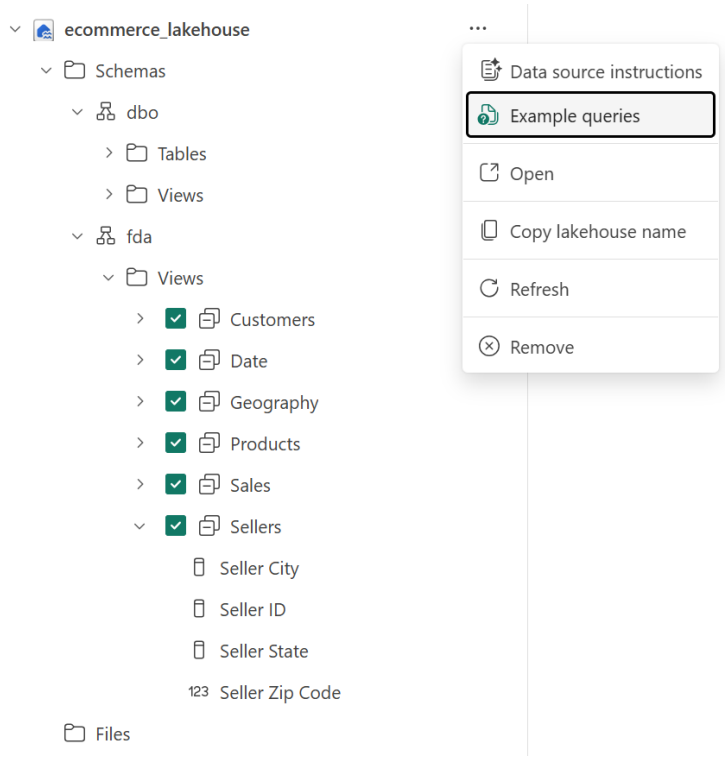
Data Agent uses the chat history as part of the context, so let’s begin with a fresh slate.

2. Ask the Question - **Which region is it most expensive to deliver to?**

3. You may see a result like below, but results may vary:



4. Let's add a Few Shot example query to help the Agent answer this question to give the Data Agent a concrete pattern to learn from. Select "Example queries" from the options, and hit the edit button:



5. Click on "Add"

6. Input the question and query pair below:

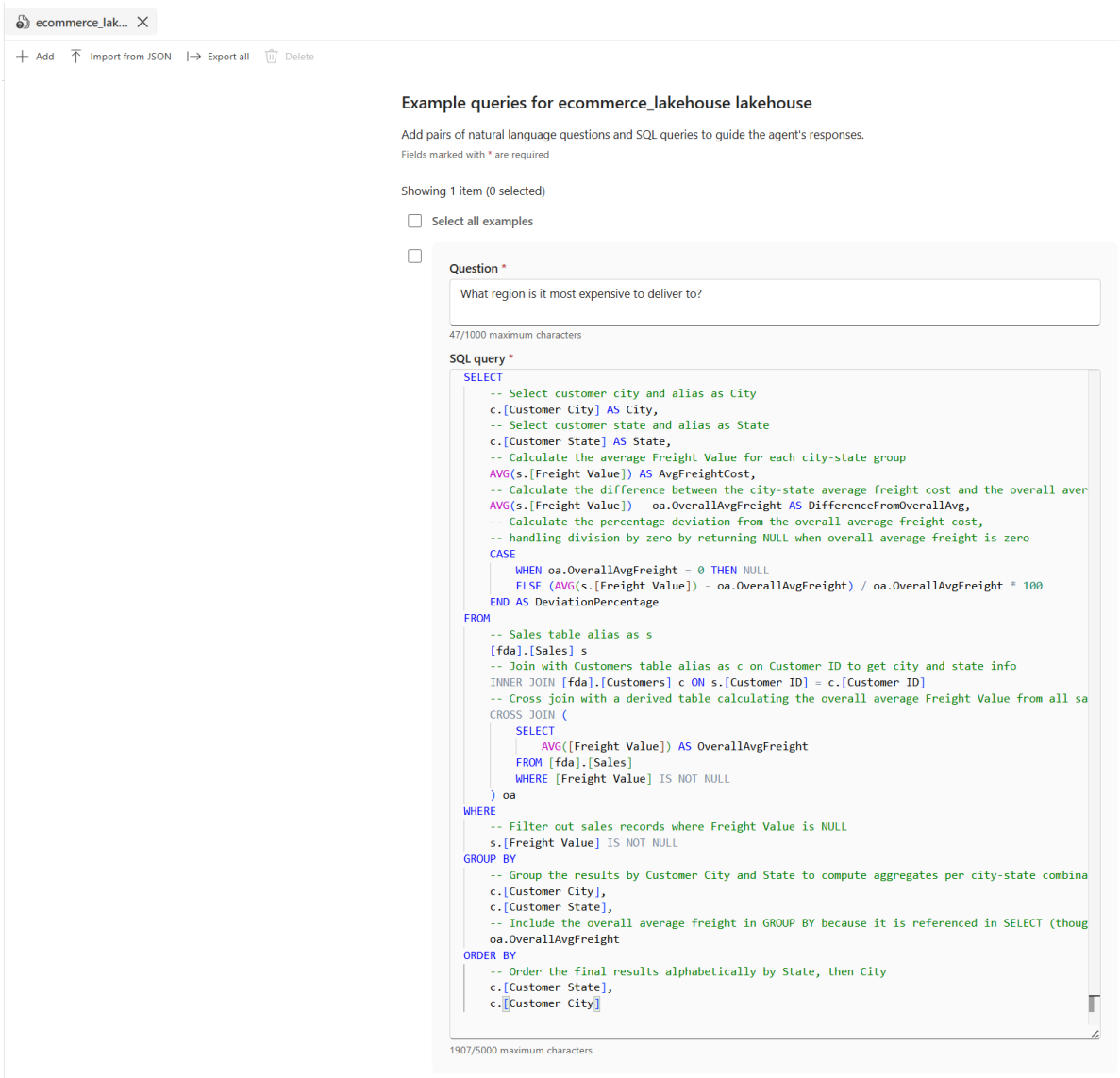
Question: What region is it most expensive to deliver to?

Query:

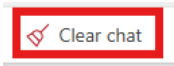
```
SELECT
  -- Select customer city and alias as City
  c.[Customer City] AS City,
  -- Select customer state and alias as State
  c.[Customer State] AS State,
  -- Calculate the average Freight Value for each city-state group
  AVG(s.[Freight Value]) AS AvgFreightCost,
```

```
   -- Calculate the difference between the city-state average freight cost and
the overall average freight cost
   AVG(s.[Freight Value]) - oa.OverallAvgFreight AS
DifferenceFromOverallAvg,
   -- Calculate the percentage deviation from the overall average freight cost,
   -- handling division by zero by returning NULL when overall average freight
is zero
   CASE
      WHEN oa.OverallAvgFreight = 0 THEN NULL
      ELSE (AVG(s.[Freight Value]) - oa.OverallAvgFreight) /
oa.OverallAvgFreight * 100
   END AS DeviationPercentage
FROM
   -- Sales table alias as s
   [fda].[Sales] s
   -- Join with Customers table alias as c on Customer ID to get city and state
info
   INNER JOIN [fda].[Customers] c ON s.[Customer ID] = c.[Customer ID]
   -- Cross join with a derived table calculating the overall average Freight
Value from all sales where the freight value is not null
   CROSS JOIN (
      SELECT
         AVG([Freight Value]) AS OverallAvgFreight
      FROM [fda].[Sales]
      WHERE [Freight Value] IS NOT NULL
   ) oa
WHERE
   -- Filter out sales records where Freight Value is NULL
   s.[Freight Value] IS NOT NULL
GROUP BY
   -- Group the results by Customer City and State to compute aggregates per
city-state combination
   c.[Customer City],
   c.[Customer State],
   -- Include the overall average freight in GROUP BY because it is referenced
in SELECT (though it will be the same for all rows)
```

```
    oa.OverallAvgFreight
ORDER BY
    -- Order the final results alphabetically by State, then City
    c.[Customer State],
    c.[Customer City]
```



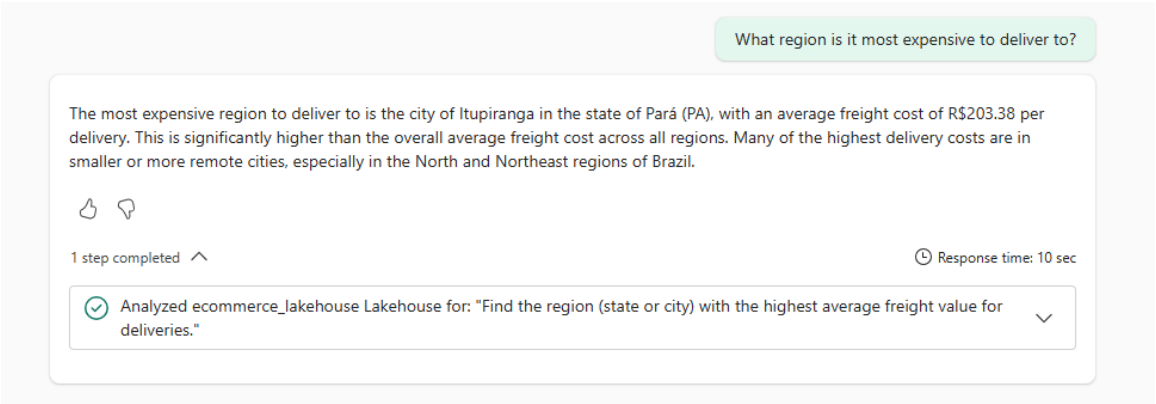
Close the window.



7. Clear the chat

   Now, ask the question again: **What region is it most expensive to deliver to?**

   Do you notice a change in the result?



As part of the run steps, you may also notice consideration of the example query:

Example queries

These example queries help guide the data agent's responses. They won't be executed directly, but are used to provide context and improve answer quality.

Question 1

What region is the most expensive to deliver to?

Sql

```sql
SELECT
    -- Select customer city and alias as City
    c.[Customer City] AS City,
    -- Select customer state and alias as State
    c.[Customer State] AS State,
    -- Calculate the average Freight Value for each city-state group
    AVG(s.[Freight Value]) AS AvgFreightCost,
    -- Calculate the difference between the city-
state average freight cost and the overall average freight cost
    AVG(s.[Freight Value]) - oa.OverallAvgFreight AS DifferenceFromOverallAvg,
    -- Calculate the percentage deviation from the overall average freight cost,
    --
 handling division by zero by returning NULL when overall average freight is zero
    CASE
        WHEN oa.OverallAvgFreight = 0 THEN NULL
        ELSE (AVG(s.[Freight Value]) -
 oa.OverallAvgFreight) / oa.OverallAvgFreight * 100
    END AS DeviationPercentage
FROM
    -- Sales table alias as s
    [fda].[Sales] s
    --
 Join with Customers table alias as c on Customer ID to get city and state info
    INNER JOIN [fda].[Customers] c ON s.[Customer ID] = c.[Customer ID]
    --
 Cross join with a derived table calculating the overall average Freight Value fro
m all sales where the freight value is not null
    CROSS JOIN (
        SELECT
            AVG([Freight Value]) AS OverallAvgFreight
        FROM [fda].[Sales]
        WHERE [Freight Value] IS NOT NULL
    ) oa
WHERE
    -- Filter out sales records where Freight Value is NULL
    s.[Freight Value] IS NOT NULL
GROUP BY
    --
 Group the results by Customer City and State to compute aggregates per city-
state combination
    c.[Customer City],
    c.[Customer State],
    --
 Include the overall average freight in GROUP BY because it is referenced in SELEC
T (though it will be the same for all rows)
    oa.OverallAvgFreight
ORDER BY
    -- Order the final results alphabetically by State, then City
    c.[Customer State],
    c.[Customer City]
```

You can also try the questions below to find out how your example queries are being applied:

- Which region has the lowest sales?
- Which region gets the slowest delivery?

Step 5: Challenge – Use what you've learned to answer additional questions

1. From what you've learned above, can you configure the Data Agent to answer the
   following questions:
   a. **Who are the top 10 sellers with the most operational inefficiencies? (Considerations: Late deliveries, time to approve orders)**
   b. **Which product categories are poorly advertised? (Considerations: Product description lengths, Product photo counts.)**

Congrats, you have officially finished the lab!